6.6 Similar Matrices

The key step in this chapter is to diagonalize a matrix by using its eigenvectors. When S is the eigenvector matrix, the diagonal matrix $S^{-1}AS$ is Λ—the eigenvalue matrix. But diagonalization is not possible for every A. Some matrices have too few eigenvectors—we had to leave them alone. In this new section, the eigenvector matrix S remains the best choice when we can find it, ***but now we allow any invertible matrix M.***

Starting from A we go to $M^{-1}AM$. This matrix may be diagonal—probably not. It still shares important properties of A. No matter which M we choose, ***the eigenvalues stay the same.*** The matrices A and $M^{-1}AM$ are called "similar". A typical matrix A is similar to a whole family of other matrices because there are so many choices of M.

DEFINITION Let M be any invertible matrix. Then $B = M^{-1}AM$ is ***similar*** to A.

If $B = M^{-1}AM$ then immediately $A = MBM^{-1}$. That means: If B is similar to A then A is similar to B. The matrix in this reverse direction is M^{-1}—just as good as M.

A diagonalizable matrix is similar to Λ. In that special case M is S. We have $A = S\Lambda S^{-1}$ and $\Lambda = S^{-1}AS$. They certainly have the same eigenvalues! This section is opening up to other similar matrices $B = M^{-1}AM$, by allowing all invertible M.

The combination $M^{-1}AM$ appears when we change variables in a differential equation. Start with an equation for $\boldsymbol{u}$ and set $\boldsymbol{u} = M\boldsymbol{v}$:

$$\frac{d\boldsymbol{u}}{dt} = A\boldsymbol{u} \quad \text{becomes} \quad M\frac{d\boldsymbol{v}}{dt} = AM\boldsymbol{v} \quad \text{which is} \quad \frac{d\boldsymbol{v}}{dt} = M^{-1}AM\boldsymbol{v}.$$

The original coefficient matrix was A, the new one at the right is $M^{-1}AM$. Changing $\boldsymbol{u}$ to $\boldsymbol{v}$ leads to a similar matrix. When $M = S$ the new system is diagonal—the maximum in simplicity. Other choices of M could make the new system triangular and easier to solve. Since we can always go back to $\boldsymbol{u}$, similar matrices must give the same growth or decay. More precisely, *the eigenvalues of A and B are the same.*

(No change in λ's) Similar matrices A and $M^{-1}AM$ have the **same eigenvalues.** If $\boldsymbol{x}$ is an eigenvector of A, then $M^{-1}\boldsymbol{x}$ is an eigenvector of $B = M^{-1}AM$.

The proof is quick, since $B = M^{-1}AM$ gives $A = MBM^{-1}$. Suppose $A\boldsymbol{x} = \lambda\boldsymbol{x}$:

$$MBM^{-1}\boldsymbol{x} = \lambda\boldsymbol{x} \quad \textit{means that} \quad B(M^{-1}\boldsymbol{x}) = \lambda(M^{-1}\boldsymbol{x}).$$

The eigenvalue of B is the same λ. The eigenvector has changed to $M^{-1}\boldsymbol{x}$.

Two matrices can have the same *repeated* λ, and fail to be similar—as we will see.

Example 1 These matrices $M^{-1}AM$ all have the same eigenvalues 1 and 0.

The projection $A = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ is similar to $\Lambda = S^{-1}AS = \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$

Now choose $M = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$. The similar matrix $M^{-1}AM$ is $\begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$.

Also choose $M = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. The similar matrix $M^{-1}AM$ is $\begin{bmatrix} .5 & -.5 \\ -.5 & .5 \end{bmatrix}$.

All 2 by 2 matrices with those eigenvalues 1 and 0 are similar to each other. The eigenvectors change with M, the eigenvalues don't change.

The eigenvalues in that example are *not repeated.* This makes life easy. Repeated eigenvalues are harder. The next example has eigenvalues 0 and 0. The zero matrix shares those eigenvalues, but it is similar only to itself: $M^{-1}0M = 0$.

Example 2 A family of similar matrices with $A = 0, 0$ (repeated eigenvalue)

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \text{ is similar to } \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \text{ and all } B = \begin{bmatrix} cd & d^2 \\ -c^2 & -cd \end{bmatrix} \text{ except } \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

These matrices B all have zero determinant (like A). They all have rank one (like A). One eigenvalue is zero and the trace is $cd - dc = 0$, so the other must be zero. I chose any $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc = 1$, and $B = M^{-1}AM$.

These matrices B can't be diagonalized. In fact A is as close to diagonal as possible. It is the "***Jordan form***" for the family of matrices B. This is the outstanding member (my class says "Godfather") of the family. The Jordan form $J = A$ is as near as we can come to diagonalizing these matrices, when there is only one eigenvector. In going from A to $B = M^{-1}AM$, some things change and some don't. Here is a table to show this.

Not changed by M	Changed by M
Eigenvalues	**Eigenvectors**
Trace and determinant	**Nullspace**
Rank	**Column space**
Number of independent eigenvectors	**Row space**
	Left nullspace
Jordan form	**Singular values**

The eigenvalues don't change for similar matrices; the eigenvectors do. The trace is the sum of the λ's (unchanged). The determinant is the product of the same λ's.[1] The nullspace consists of the eigenvectors for $\lambda = 0$ (if any), so it can change. Its dimension $n - r$ does not change! The *number* of eigenvectors stays the same for each λ, while the vectors themselves are multiplied by M^{-1}. The *singular values* depend on $A^{\mathrm{T}}A$, which definitely changes. They come in the next section.

[1] The determinant is unchanged because $\det B = (\det M^{-1})(\det A)(\det M) = \det A$.

Examples of the Jordan Form

The *Jordan form* is the serious new idea here. We lead up to it with one more example of similar matrices: *triple eigenvalue, one eigenvector.*

Example 3 This **Jordan matrix** J has $\lambda = 5, 5, 5$ on its diagonal. Its only eigenvectors are multiples of $\boldsymbol{x} = (1, 0, 0)$. Algebraic multiplicity is 3, geometric multiplicity is 1:

$$\text{If } J = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \text{ then } J - 5I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ has rank 2.}$$

Every similar matrix $B = M^{-1}JM$ has the same triple eigenvalue $5, 5, 5$. Also $B - 5I$ must have the same rank 2. Its nullspace has dimension 1. So every B that is similar to this "Jordan block" J has only one independent eigenvector $M^{-1}\boldsymbol{x}$.

The transpose matrix J^{T} has the same eigenvalues $5, 5, 5$, and $J^{\mathrm{T}} - 5I$ has the same rank 2. ***Jordan's theorem says that J^{T} is similar to J.*** The matrix M that produces the similarity happens to be the *reverse identity*:

$$J^{\mathrm{T}} = M^{-1}JM \quad \text{is} \quad \begin{bmatrix} 5 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 5 \end{bmatrix} = \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix} \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix}.$$

All blank entries are zero. An eigenvector of J^{T} is $M^{-1}(1, 0, 0) = (0, 0, 1)$. There is one line of eigenvectors $(x_1, 0, 0)$ for J and another line $(0, 0, x_3)$ for J^{T}.

The key fact is that this matrix J is similar to *every* matrix A with eigenvalues $5, 5, 5$ and *one line* of eigenvectors. There is an M with $M^{-1}AM = J$.

Example 4 Since J is as close to diagonal as we can get, the equation $d\boldsymbol{u}/dt = J\boldsymbol{u}$ cannot be simplified by changing variables. We must solve it as it stands:

$$\frac{d\boldsymbol{u}}{dt} = J\boldsymbol{u} = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{is} \quad \begin{aligned} dx/dt &= 5x + y \\ dy/dt &= 5y + z \\ dz/dt &= 5z. \end{aligned}$$

The system is triangular. We think naturally of back substitution. Solve the last equation and work upwards. Main point: *All solutions contain* e^{5t} since $\lambda = 5$:

Last equation	$\dfrac{dz}{dt} = 5z$	yields	$z = z(0)e^{5t}$
Notice te^{5t}	$\dfrac{dy}{dt} = 5y + z$	yields	$y = \big(y(0) + tz(0)\big)e^{5t}$
Notice t^2e^{5t}	$\dfrac{dx}{dt} = 5x + y$	yields	$x = \big(x(0) + ty(0) + \frac{1}{2}t^2z(0)\big)e^{5t}$.

The two missing eigenvectors are responsible for the te^{5t} and t^2e^{5t} terms in y and x. The factors t and t^2 enter because $\lambda = 5$ is a triple eigenvalue with one eigenvector.

Note Chapter 7 will explain another approach to similar matrices. Instead of changing variables by $\boldsymbol{u} = M\boldsymbol{v}$, we "***change the basis***". In this approach, similar matrices will represent the same transformation of n-dimensional space. When we choose a basis for $\mathbf{R}^n$, we get a matrix. The standard basis vectors ($M = I$) lead to $I^{-1}AI$ which is A. Other bases lead to similar matrices $B = M^{-1}AM$.

The Jordan Form

For every A, we want to choose M so that $M^{-1}AM$ is as ***nearly diagonal as possible.*** When A has a full set of n eigenvectors, they go into the columns of M. Then $M = S$. The matrix $S^{-1}AS$ is diagonal, period. This matrix Λ is the Jordan form of A—when A can be diagonalized. In the general case, eigenvectors are missing and Λ can't be reached.

Suppose A has s independent eigenvectors. Then it is similar to a matrix with s blocks. Each block is like J in Example 3. *The eigenvalue is on the diagonal with* 1's *just above it.* This block accounts for one eigenvector of A. When there are n eigenvectors and n blocks, they are all 1 by 1. In that case J is Λ.

(Jordan form) If A has s independent eigenvectors, it is similar to a matrix J that has s Jordan blocks on its diagonal: Some matrix M puts A into Jordan form:

Jordan form
$$M^{-1}AM = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_s \end{bmatrix} = J. \tag{1}$$

Each block in J has one eigenvalue λ_i, one eigenvector, and 1's above the diagonal:

Jordan block
$$J_i = \begin{bmatrix} \lambda_i & 1 & \\ & \ddots & \\ & & 1 \\ & & \lambda_i \end{bmatrix}. \tag{2}$$

A is similar to B if they share the same Jordan form J—not otherwise.

The Jordan form J has an off-diagonal 1 for each missing eigenvector (and the 1's are next to the eigenvalues). This is the big theorem about matrix similarity. In every family of similar matrices, we are picking one outstanding member called J. It is nearly diagonal (or if possible completely diagonal). For that J, we can solve $d\boldsymbol{u}/dt = J\boldsymbol{u}$ as in Example 4. We can take powers J^k as in Problems 9–10. Every other matrix in the family has the form $A = MJM^{-1}$. The connection through M solves $d\boldsymbol{u}/dt = A\boldsymbol{u}$.

The point you must see is that $MJM^{-1}MJM^{-1} = MJ^2M^{-1}$. That cancellation of $M^{-1}M$ in the middle has been used through this chapter (when M was S). We found A^{100} from $S\Lambda^{100}S^{-1}$—by diagonalizing the matrix. Now we can't quite diagonalize A. So we use $MJ^{100}M^{-1}$ instead.

Jordan's Theorem is proved in my textbook *Linear Algebra and Its Applications*. Please refer to that book (or more advanced books) for the proof. The reasoning is rather intricate and in actual computations the Jordan form is not at all popular—its calculation is not stable. A slight change in A will separate the repeated eigenvalues and remove the off-diagonal 1's—switching to a diagonal Λ.

Proved or not, you have caught the central idea of similarity—to make A as simple as possible while preserving its essential properties.

■ REVIEW OF THE KEY IDEAS ■

1. B is similar to A if $B = M^{-1}AM$, for some invertible matrix M.
2. Similar matrices have the same eigenvalues. Eigenvectors are multiplied by M^{-1}.
3. If A has n independent eigenvectors then A is similar to Λ (take $M = S$).
4. Every matrix is similar to a Jordan matrix J (which has Λ as its diagonal part). J has a block for each eigenvector, and 1's for missing eigenvectors.

■ WORKED EXAMPLES ■

6.6 A The 4 by 4 triangular Pascal matrix A and its inverse (alternating diagonals) are

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}.$$

Check that A and A^{-1} have the same eigenvalues. Find a diagonal matrix D with alternating signs that gives $A^{-1} = D^{-1}AD$. This A is similar to A^{-1}, which is unusual.

These similar matrices must have the same Jordan form J. This J has only one block because the Pascal matrix has only one line of eigenvectors.

Solution The triangular matrices A and A^{-1} both have $\lambda = 1, 1, 1, 1$ on their main diagonals. Choose D with alternating 1 and -1 on its diagonal. D *equals* D^{-1}:

$$D^{-1}AD = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & -1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & -1 & \\ & & & 1 \end{bmatrix} = A^{-1}.$$

Check: Changing signs in rows 1 and 3 of A, and columns 1 and 3, produces the four negative entries in A^{-1}. We are multiplying row i by $(-1)^i$ and column j by $(-1)^j$, which gives the alternating diagonals in A^{-1}. Then AD has *columns with alternating signs.*

6.6 B ***The best way to compute eigenvalues of a large matrix is not from solving*** $\det(A - \lambda I) = 0$. That high degree polynomial is a numerical disaster.

Instead we compute similar matrices $A_1, A_2, \ldots$ that approach a triangular matrix. Then the eigenvalues of A (unchanged) are almost sitting on the main diagonal.

One way is to factor $A = QR$ by "Gram-Schmidt". Reverse the order to $A_1 = RQ$. This matrix is similar to A because $RQ = Q^{-1}(QR)Q$. An example with $c = \cos\theta$ and $s = \sin\theta$ shows how a small off-diagonal s can be *cubed* in A_1:

$$A = \begin{bmatrix} c & s \\ s & 0 \end{bmatrix} \text{ factors into } \begin{bmatrix} c & s \\ s & -c \end{bmatrix}\begin{bmatrix} 1 & cs \\ 0 & s^2 \end{bmatrix} = QR.$$

$$A_1 = RQ = \begin{bmatrix} c + cs^2 & s^3 \\ s^3 & -cs^2 \end{bmatrix} \text{ has } s^3 \text{ below the diagonal}$$

Another step can factor $A_1 = Q_1R_1$ and reverse to $A_2 = R_1Q_1$. This ***QR method*** is in Section 9.3 with a further improvement for A_1. Add cs^2 to its diagonal (to get zero in the corner) and then subtract back from A_2:

Shift and factor $A_1 + cs^2I = Q_1R_1$ **Reverse and shift back** $A_2 = R_1Q_1 - cs^2I$

Shifted QR is an amazing success—just about the best way to compute eigenvalues.

Problem Set 6.6

1 If $C = F^{-1}AF$ and also $C = G^{-1}BG$, what matrix M gives $B = M^{-1}AM$? Conclusion: If C is similar to A and also to B then _____.

2 If $A = \text{diag}(1, 3)$ and $B = \text{diag}(3, 1)$ show that A and B are similar (find an M).

3 Show that A and B are similar by finding M so that $B = M^{-1}AM$:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}.$$

4 If a 2 by 2 matrix A has eigenvalues 0 and 1, why is it similar to $\Lambda = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$? Deduce from Problem 1 that all 2 by 2 matrices with those eigenvalues are similar.

5 Which of these six matrices are similar? Check their eigenvalues.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

6 There are sixteen 2 by 2 matrices whose entries are 0's and 1's. Similar matrices go into the same family. How many families? How many matrices (total 16) in each family?

7 (a) If x is in the nullspace of A show that $M^{-1}x$ is in the nullspace of $M^{-1}AM$.

(b) The nullspaces of A and $M^{-1}AM$ have the same (vectors)(basis)(dimension).

8 Suppose $Ax = \lambda x$ and $Bx = \lambda x$ with the same λ's and x's. With n independent eigenvectors we have $A = B$: *Why*? Find $A \neq B$ when both have eigenvalues $0, 0$ but only one line of eigenvectors $(x_1, 0)$.

9 By direct multiplication find A^2 and A^3 and A^5 when

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Guess the form of A^k. Set $k = 0$ to find A^0 and $k = -1$ to find A^{-1}.

Questions 10–14 are about the Jordan form.

10 By direct multiplication, find J^2 and J^3 when

$$J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}.$$

Guess the form of J^k. Set $k = 0$ to find J^0. Set $k = -1$ to find J^{-1}.

11 Solve $du/dt = Ju$ for J in Problem 10, starting from $u(0) = (5, 2)$. Remember $te^{\lambda t}$.

12 These Jordan matrices have eigenvalues $0, 0, 0, 0$. They have two eigenvectors (one from each block). But the block sizes don't match and they are *not similar*:

$$J = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{and} \quad K = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right].$$

For any matrix M, compare JM with MK. If they are equal show that M is not invertible. Then $M^{-1}JM = K$ is impossible: *J is not similar to K.*

13 Based on Problem 12, what are the five Jordan forms when $\lambda = 0, 0, 0, 0$?

14 Prove that A^{T} *is always similar to* A (we know the λ's are the same):

1. For one Jordan block J_i: Find M_i so that $M_i^{-1}J_iM_i = J_i^{\mathrm{T}}$ (see Example 3).
2. For any J with blocks J_i: Build M_0 from blocks so that $M_0^{-1}JM_0 = J^{\mathrm{T}}$.
3. For any $A = MJM^{-1}$: Show that A^{T} is similar to J^{T} and so to J and to A.

15 Prove that $\det(A - \lambda I) = \det(M^{-1}AM - \lambda I)$. (You could write $I = M^{-1}M$ and factor out $\det M^{-1}$ and $\det M$.) Since these *characteristic polynomials* of A and $M^{-1}AM$ are the same, the eigenvalues are the same (with the same multiplicities).

16 Which pairs are similar? Choose a, b, c, d to prove that the other pairs aren't:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \begin{bmatrix} b & a \\ d & c \end{bmatrix} \quad \begin{bmatrix} c & d \\ a & b \end{bmatrix} \quad \begin{bmatrix} d & c \\ b & a \end{bmatrix}.$$

17 True or false, with a good reason:

(a) A symmetric matrix can't be similar to a nonsymmetric matrix.

(b) An invertible matrix can't be similar to a singular matrix.

(c) A can't be similar to $-A$ unless $A = 0$.

(d) A can't be similar to $A + I$.

18 If B is invertible, prove that AB is similar to BA. *They have the same eigenvalues.*

19 If A is 6 by 4 and B is 4 by 6, AB and BA have different sizes. But with blocks

$$M^{-1}FM = \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix} = G.$$

(a) What sizes are the four blocks (the same four sizes in each matrix)?

(b) This equation is $M^{-1}FM = G$, so F and G have the same 10 eigenvalues. F has the 6 eigenvalues of AB plus 4 zeros; G has the 4 eigenvalues of BA plus 6 zeros. ***AB has the same eigenvalues as BA plus _____ zeros.***

20 Why are these statements all true?

(a) If A is similar to B then A^2 is similar to B^2.

(b) A^2 and B^2 can be similar when A and B are not similar (try $\lambda = 0, 0$).

(c) $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ is similar to $\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$.

(d) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ is not similar to $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.

(e) If we exchange rows 1 and 2 of A, and then exchange columns 1 and 2, **the eigenvalues stay the same.** In this case $M =$ _____.

21 If J is the 5 by 5 Jordan block with $\lambda = 0$, find J^2 and count its eigenvectors and find its Jordan form (there will be two blocks).

Challenge Problems

22 If an n by n matrix A has all eigenvalues $\lambda = 0$, prove that $A^n =$ zero matrix. (Maybe prove first that $J^n =$ zero matrix, by direct multiplication. Or use the Cayley-Hamilton Theorem?)

23 For the shifted QR method in the Worked Example **6.6 B**, show that A_2 is similar to A_1. No change in eigenvalues, and the A's quickly approach a diagonal matrix.

24 If A is similar to A^{-1}, must all the eigenvalues equal 1 or -1?